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بوتکمپ

Artificial Intelligence

MLOps

Session 7

Session Overview

Recap: Packaging, deployment, and security

Today's Journey: Production Operations & Monitoring:

- Production deployment strategies
- Monitoring vs observability concepts
- Model performance degradation and drift
- Logging strategies for ML applications
- Incident response and troubleshooting
- Basic alerting and notification systems

VisionaryAI Production Systems

Real Production ML Systems

Computer Vision: Mobile phone factory defect detection

- 1,000 images/minute processing
- <50ms latency requirement
- Production line halt = \$1000/minute cost

NLP: AI helpdesk agent

- 500 tickets/hour processing
- Direct customer satisfaction impact
- Real-time response suggestions

Recommender: E-commerce product recommendations

- 10,000 concurrent users
- Immediate conversion rate impact
- Revenue-critical system

Production vs Development

Why Production is Different

Development Environment:

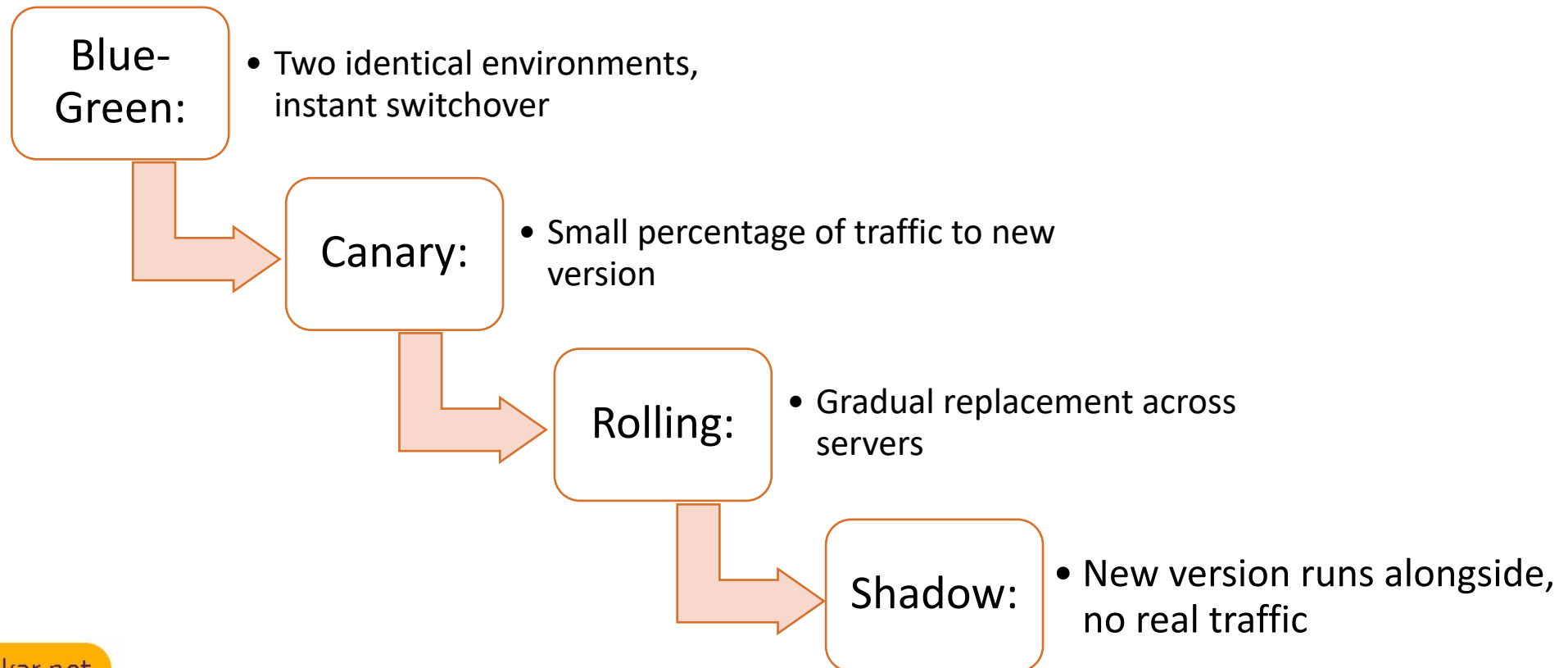
- Clean, curated data
- Known edge cases
- Forgiving latency requirements
- Easy rollbacks

Production Environment:

- Real, messy data
- Unknown edge cases
- Strict SLA requirements
- Business impact of failures

Deployment Strategies Overview

Production Deployment Strategies

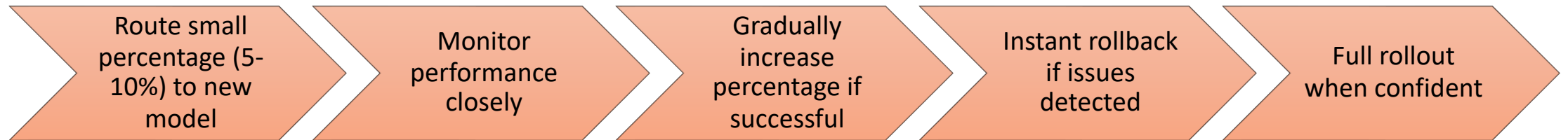


Blue-Green Deployment



Use Case: Major model updates with zero-downtime requirement

Canary Deployment



Rolling & Shadow Deployment

Rolling Deployment:

- Replace servers one at a time
- Gradual transition
- Lower risk than big-bang

Shadow Deployment:

- New model runs in parallel
- No impact on production traffic
- Compare predictions safely
- Risk-free testing

Monitoring vs Observability

Monitoring (Reactive):

- Known metrics and thresholds
- Alerts on predefined conditions
- "What happened?"
- Dashboard-driven

Observability (Proactive):

- Explore unknown problems
- Trace through system behavior
- "Why did this happen?"
- Investigation-driven

ML System Monitoring Metrics

What to Monitor in ML Systems:

1. Business Metrics

- Conversion rates, customer satisfaction
- Revenue impact, user engagement

2. Model Performance Metrics

- Accuracy, precision, recall
- Prediction confidence distributions

3. System Performance Metrics

- Latency, throughput, error rates
- CPU/memory/GPU utilization

4. Data Quality Metrics

- Completeness, freshness, schema violations
- Feature distribution changes

Model Drift Overview

Model Drift - The Silent Killer

- **Traditional Software:** If $2+2=4$ today, still true tomorrow
- **ML Models:** Learn from past data, assume future similarity
- **When assumptions break:** Models drift and degrade

Key Insight: Your server can be perfectly healthy while your model is completely broken

Types of Model Drift

1. Data Drift (Covariate Shift)

- Input data characteristics change
- Relationship stays same
- Example: Camera upgrade changes image quality

2. Concept Drift

- Input-output relationship changes
- Example: Fashion trends change accessory preferences

3. Label Drift

- Target class distribution changes
- Example: Shift from billing to return tickets

Real-World Drift Example

COVID-19: Drift at Global Scale

- **Pre-2020:** Recommendation models trained on "normal" behavior
- **2020:** Global behavior shift
 - Party clothes → Home office equipment
 - Travel → Streaming subscriptions
 - Restaurants → Grocery delivery
- **Result:** Unprecedented concept drift
- **Lesson:** External events can invalidate models overnight

Detecting Model Drift

How to Detect Model Drift

Statistical Methods:

- Kolmogorov-Smirnov test (continuous features)
- Chi-square test (categorical features)
- Compare training vs production data distributions

Performance-Based Detection:

- Monitor accuracy, precision, recall trends
- Track business metric degradation

Prediction Distribution Monitoring:

- Confidence score distributions
- Class prediction frequencies
- Prediction uncertainty patterns

Handling Model Drift

Retraining

- Collect new labeled data
- Retrain model periodically
- Full model replacement

Online Learning

- Continuously update with new data
- Incremental model updates
- Real-time adaptation

Ensemble Methods

- Multiple models from different time periods
- Weighted predictions based on recency

Human-in-the-Loop

- Flag uncertain predictions
- Expert review and correction
- Active learning approach

ML Logging Strategy

What to Log in ML Applications



Input Data Logs

Raw inputs,
preprocessed
features

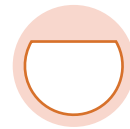
Data quality
flags, missing
values



Model Prediction Logs

Raw outputs,
post-processed
predictions

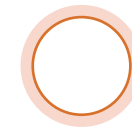
Confidence
scores, model
version



Feedback & Ground Truth

Actual outcomes,
user feedback

Human
corrections



System Performance

Response times,
error rates

Resource usage
metrics

Logging Example

VisionaryAI Logging Examples

Defect Detection Log:

```
{  
  "timestamp": "2024-03-15T10:30:45Z",  
  "image_id": "IMG_12345",  
  "image_size": [1920, 1080],  
  "brightness": 0.65,  
  "processing_time": 0.023,  
  "quality_flags": ["low_contrast"]  
}
```

Helpdesk Agent Log:

```
{  
  "ticket_id": "TKT_67890",  
  "model_version": "nlp_v2.1.3",  
  "predicted_category": "billing_issue",  
  "confidence": 0.87,  
  "processing_time": 0.15  
}
```

Logging Best Practices

- ❑ **Structured Logging:** Use JSON, not plain text
- ❑ **Correlation IDs:** Trace requests end-to-end
- ❑ **Appropriate Log Levels:** DEBUG, INFO, WARNING, ERROR, CRITICAL
- ❑ **Privacy Considerations:** Log features, not raw PI
- ❑ **Storage & Retention:** Plan for volume and lifecycle

Warning: ML logs can be massive - plan storage accordingly

Common ML Production Incidents

When Things Go Wrong

Silent Failures

System appears healthy, model quality degraded
Business metrics tank, system metrics normal

Performance Degradation

Model too slow for production SLAs
Resource contention issues

Data Pipeline Failures

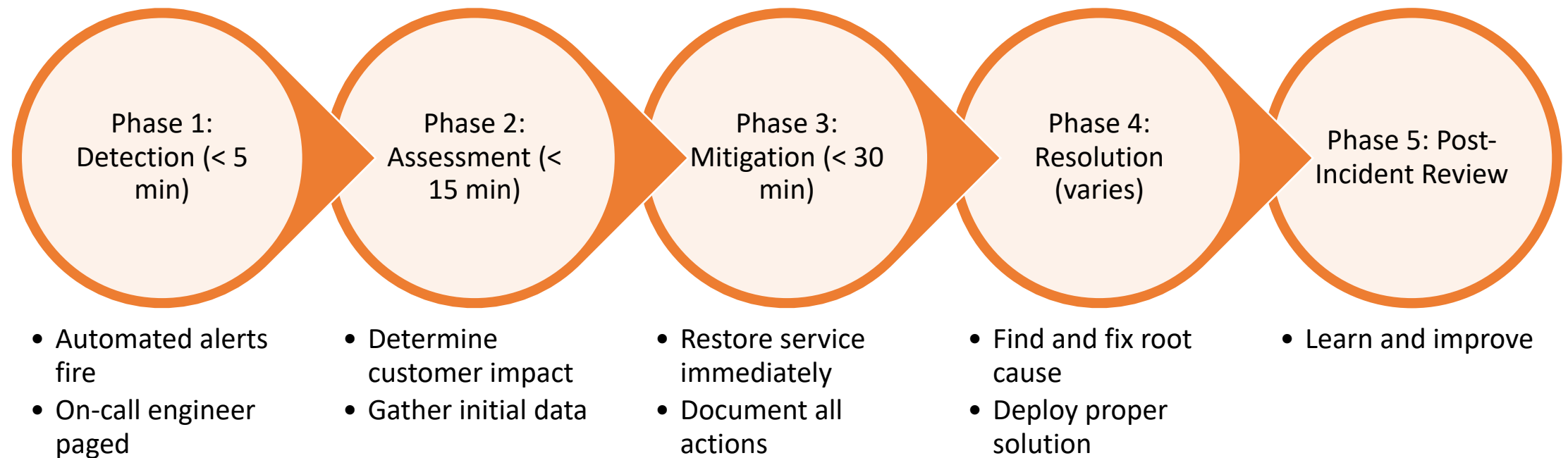
Schema changes break feature extraction
New categories not in training data

Model Serving Failures

Infrastructure crashes, OOM errors
Network connectivity problems

Incident Response Framework

ML Incident Response Process



Real Incident Example

VisionaryAI Incident Case Study

- **Problem:** Customer support AI categorizing all tickets as "billing_issue"
- **Detection:** Alert fired - category distribution shifted dramatically
- **Assessment:** High confidence scores, but single category output
- **Mitigation:** Route to humans, disable AI suggestions
- **Resolution:** NLP library update changed tokenization
- **Prevention:** Integration tests for library updates
- **Lesson:** External dependencies can break ML systems silently

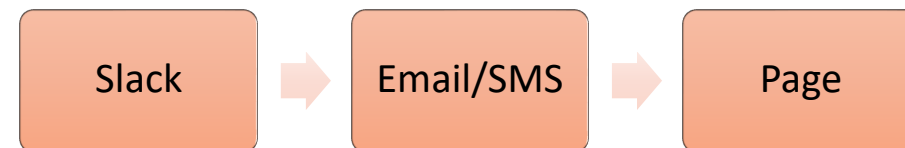
ML System Alerts

Alert Types:

1. Threshold-Based: Simple rules (accuracy < 90%)
2. Anomaly-Based: Statistical deviations
3. Business Impact: Focus on what matters

Best Practices:

- Avoid alert fatigue
- Make alerts actionable
- Use escalation policies
- Provide context in alerts



Example Alert Setup

VisionaryAI Alert Configuration

Warning Level (Slack):

- Model accuracy: 94-90%
- Response time: 50-100ms
- Error rate: 1-3%

Critical Level (PagerDuty):

- Model accuracy: <90%
- Response time: >100ms
- Error rate: >3%

Executive Dashboard:

- Business KPI impacts
- Revenue implications
- Customer satisfaction trends

Key Takeaways

Session Summary

- ✓ Production requires monitoring both infrastructure AND model intelligence
- ✓ Choose deployment strategy based on risk tolerance
- ✓ Monitor known problems, observe unknown ones
- ✓ Model drift is inevitable - plan for it
- ✓ Log comprehensively for debugging and analysis
- ✓ Have incident response plan before you need it
- ✓ Smart alerting prevents customer-facing issues

Next Session Preview

Coming Up: CI/CD & MLOps Best Practices

Session 8 Preview:

- Continuous Integration for ML projects
- Automated testing and validation pipelines
- Deployment automation
- Team collaboration workflows
- Career paths in MLOps

How it connects:

- Automate the processes we learned today

Questions & Discussion

- Questions about production operations?
- Real-world monitoring experiences?
- Challenges with model drift?
- Incident response stories?

Code Demo: Monitoring dashboard examples and log analysis

دانشکار

THANK YOU

FOR YOUR ATTENTION!

